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An Introduction of Novel Additive to Enhance the Performance of the Slick Water Formulation to Improve Phase Behavior and Regain Permeability

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Abstract

Slick water fracturing fluid is characterized by the incorporation of a specified quantity of proppants and other additives, including surfactants, friction-reducing agents, and anti-swelling agents in the aqueous medium. Since the early 1950s, slickwater has been employed in the development of oil and gas reservoirs. The primary method for hydraulic fracturing of unconventional shale reservoirs involves the use of slick water formulations. The efficacy of slick water is evaluated based on the concentration of friction reducers and other chemicals, and their influence on viscosity, surface tension, aging tests, and regain permeability tests. It is essential to ensure a clean and clear phase behavior of high viscosity friction reducers (HVFR) formulations to reduce damage to regain permeability.

A typical high viscosity friction reducer (HVFR) formulation consists of friction reducer polymer, surfactant, biocide, clay stabilizer, and breaker, all incorporated into water to achieve the desired viscosity, surface tension, and regain permeability specifications. In certain brine compositions that contains significant amounts of divalent cations HVFR formulations may become unstable in high temperature applications. In such cases, adjustments to the formulations are necessary, and new additives may need to be introduced to maintain a clean and effective breaking of the slick water. Key variables affecting the performance of the system include surface tension, viscosity, mixed micellar formation, rate of polymer scission, and deflocculation of polymer particles.

In this study, we propose a novel additive that preferentially migrates to the interface, effectively reducing the surface tension below 25 mN/m. The conventional system, composed of a friction reducer, surfactant, clay stabilizer, biocide, and breaker, demonstrated unstable phase behavior after aging the sample for 24 hours at 270°F. Root cause analysis, screening, and elimination methodologies were employed to identify the performance issue in the HVFR slurry. The inclusion of the novel additive enhanced the performance of primary surfactants and prevented the broken friction reducer polymer from flocculating, thereby keeping it well suspended. Additionally, the surface tension of the entire formulation was reduced by up to 5 mN/m with the addition of one-tenth of the novel additive concentration compared to the primary surfactant. The

reduced surface tension enhances the penetration of the HVFR in the formation as well as retrieval of the fluid during the flow back, in turn giving less permeability damage.

This study introduced a novel additive to enhance mixed micelle formation, improving polymer phase behavior and HVFR flowability in porous media. The additive reduced permeability damage and increased regain permeability by up to 7% compared to conventional HVFR formulations. This high-temperature HVFR formulation minimized phase separation of the friction reducer polymer after 24-hour aging and kept broken polymer particles suspended. Overall, the novel additive significantly improved HVFR performance for high-temperature and high-salinity slick water applications.

Introduction

In unconventional shale reservoirs Low porosity and permeability formation are one of the largest issues faced during hydrocarbon extraction, the solution is hydraulic fracturing to increase the porosity and permeability through hydraulic fracturing by pumping sand using carrying fluid, this carrying fluid is called high viscosity friction reducers (HVFR). HVFR is used for many other purposes not just carrying the sand, but most importantly reduce the friction from the pumping at high rates and allowing to reach higher pressure the bottom hole pressure to include the fractures (AL-Hulail et al, 2021).

Friction reducer is achieved through a group of chemicals that goes in a slick water, where, depending on the total dissolved solids (TDS) the FR might lose its stability under high temperatures, surfactant, biocide, clay stabilizer, and breaker, all incorporated into water to achieve the desired viscosity. The functional tests that are critical for HVFR formulations are the measurement of surface tension, the aging test, and the regain permeability test. The aging test is critical to know the phase behavior of the polymer after it is broken, which causes partial blockage of the pore system. The objective is to have high regain permeability after the slick water application.

The objective of designing a typical slick water mixture is essentially to create a carefully balanced cocktail of five key ingredients: water, friction reducer, biocide, clay stabilizer, and surfactant. The quality of water is critical for the performance of the friction reducer. Some friction reducers are tougher than others when it comes to handling hard brine conditions, which is why choosing the optimized formulation for your specific water chemistry is crucial. Biocides also play a critical role. Without biocide, high polymer degradation might happen, which will impact the friction reduction performance. Clay stabilizers prevent swelling of the clay when it meets water. This functionality is critical for the performance of the HVFR fluid system. The surfactant plays a critical role in reducing the surface tension of water. Research by Kotb and colleagues in 2022, along with Ali's team in 2020, shows that surfactants also help the entire mixture tolerate high-salt environments much better. When it comes time for flowback, surfactants with low surface tension are needed (Qi et al., 2023).

Current technologies consist of polymers with backbones of C-C proved to be challenging to degrade naturally this along with not considering the proper oxidizer carefully from reservoir conditions and data it can cause damage to the conductivity (Vazquez et al, 2022). The introduction of the novel additive adds to the performance of primary surfactants and most importantly it will prevent the broken polymer from flocculating and causing the aforementioned damage to the formation, thereby keeping it well suspended. Additionally, the surface tension of the entire formulation was reduced by up to 5 mN/m with the addition of one-tenth of the novel additive concentration compared to the primary surfactant.

Ali's team in 2020 studied the impact of the chemistry of completions fluids on effective permeability and when studying the effect of varying completions fluids, surfactant choice in the fluid is rather crucial in system selection. The surfactant selection is critical, for similar organic shale reservoir at two different depths using two different surfactants, one had regained permeability ranging between 50-85%, while other has regain permeability of less than 20%. This clearly demonstrates that the surfactant chemistry and its interaction with polymer is critical for the slick water application.

This novel additive forms a mixed micelles between alcohol ethoxylate and silicone polyether (a performance enhancer) to enhances surface properties and improves the phase behavior of the polymer by complementing to the already existence surfactant. The surfactant that is present in the in slickwater has larger conformance volume consisting of a hydrophilic head, and hydrophobic tail (figure #1 left). This arrangement increases the FR aggregation which impacts the regain permeability of the slick water, this is where this novel additive comes into play where it fills the empty spaces between the already existent surfactant(figure #1 right), and help aid in Increase in temperature tolerance, Lowers surface tension, and Reduces agglomeration of FR in the regain permeability test by lowering the aggregation in the mix significantly in turn increases return perm. The smaller chain silicone polyether that does not take part in formation micelle, interact with polymer to avoid any aggregation of the polymer.

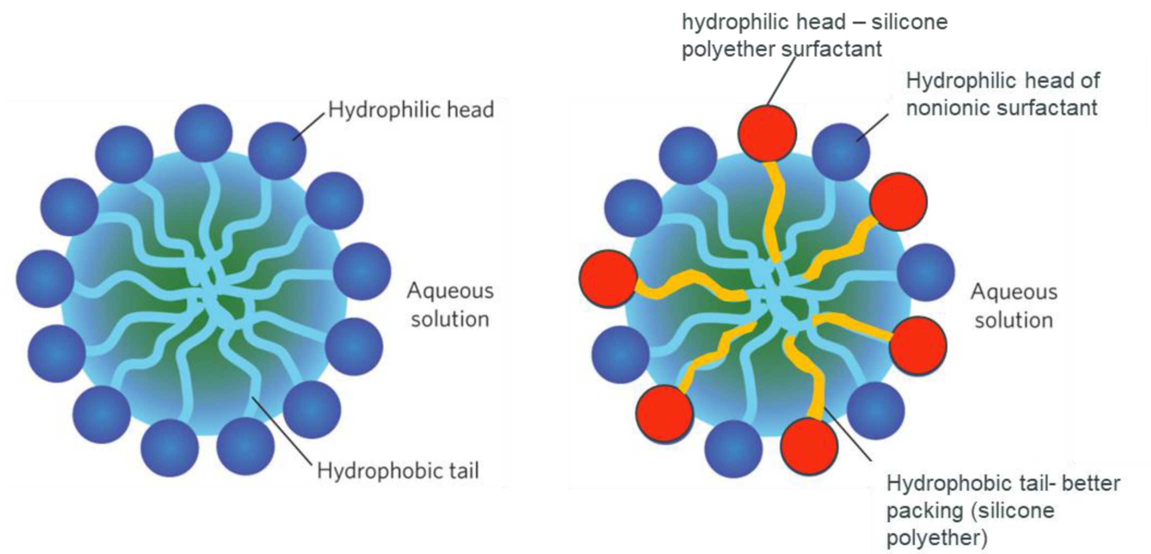


Figure 1—Surfactant head and tail arrangement (w/o additive on left) & (w/ novel additive on right)

This study, we propose a novel additive that uses two types of surfactants, one is nonionic and the other is Silicone polyether based preferentially migrates to the interface, effectively reducing the surface tension below 25 mN/m and improve regain permeability damage by 10 % than compared to conventional HVFR formulations. Not only that, the novel additive enhanced mixed micelle formation, by improving polymer phase behavior and HVFR flowability in porous media. This high-temperature HVFR formulation minimized phase separation of the friction reducer polymer after 24-hour aging and kept broken polymer particles suspended. Overall, the novel additive significantly improved HVFR performance for high temperature and high salinity slick water applications.

Methodology

This part will cover a variety of tests that were conducted for the evaluation, starting from routine test for the slickwater such as viscosity, and flow loop that give us the amount of friction reduction the FR is able to achieve, these test are carried out in order to explore the operational factors where the viscosity are used to calculate the sand carrying capacity for the sand or proppant using Reynolds number equation that is essential in pumping FR, and for the flow loop friction reduction is important while pumping to reduce the friction pressure inside the lines, more in-depth tests are carried out to explore expansive properties of the slick water, like what happens during shut in period and how it reacts during it, and also for the flow back to knowing the properties of the broken fluid is also essential such as interfacial tension and effectiveness of surfactant during flowback, the test carried out in this study are Interfacial Tension (IFT) in order to measure the impact on the surface tension and to test the theory we have illustrated previously, Phase behavior through aging after aging the sample for 24 hours at 270°F to study aggregate formation and their behavior and impact on regain permeability. The loss in regain permeability might cause a potential damage to the system, this test is done using a high precision pump to flow the slick water through a core plug simulating the treatment that is done in the formation.

Slick water data:

The system has been mixed with different loading of both the FR and the additive, to build a correlation between the concentration and viscosity and effectiveness in the IFT

Slick water mixture tested

Table 1—Slick water mixture information

Chemical	Loading (gpt)	Amount in mL	notes
Water		400	
Friction reducer	x	x	Different loadings tested
Biocide	0.25	0.1	
Clay Stabilizer	0.5	0.2	
Surfactant	1	0.4	
Special Additive	x	x	Different loadings tested
Breaker	x	x	Scales with Friction reducer

Mixing procedure

We have used following mixing procedures to observe if the mixing procedure has any effect on the viscosity of the slick water:

We start by adding the water in a constant speed mixer at 2000 rpm and we add the powder breaker, then we add the required amount of the FR and let it mix for 3 minutes, after that we lower the rpm to 1600 rpm and start adding the rest of the additives in the following order (biocide, clay stabilizer, and surfactant) and keep it mixing for 30 seconds.

Viscosity

Viscosity is measured using fan 35 atmospheric viscometer, 2 types of measurement is taken, the first is through different rpm (100, 200, 300) and readings taken when stable at each rpm, then a new mix is prepared and measured at different intervals at a constant 300 rpm (30sec, 1min, 2min, 3min).

Flow loop

Mini-Loop instrument MODEL 6500-M from Chandler engineering is used for the friction reduction efficiency testing using 13 L of Synthetic water and then add all the additives, we let the chemicals mix testing making sure that all additives mixed properly and then zero all gauges to have a baseline measurement to see just the effect of FR, then we start the schedule and add the FR after 2 minutes of the schedule start, we then collect the date after the test conclude.

Aging tests

This test is carried out through an aging cell and placed in the oven for 24 hours, new mix is prepared, the mix is poured in a 250 mL bottle and before picture is taken for comparison and then the bottle is placed in the aging cell, then the aging cell is to be pressurized using nitrogen to 500 psi and then its placed in the oven at 275°F for 24 hours, later the cell is to be cooled to room temperature, the cell is depressurized and the bottle extracted and photographed.

Regain permeability

A core plug sample is used to measure the regained permeability using our FR, the sample is to be saturated using 3% KCl brine through vacuum assisted imbibition for at least 3 hrs, during the saturation NMR Scans are used to identify saturation level, after the sample is saturated the sample will be loaded in hassler style core holder pressure vessel, where both axial and radial pressure to simulate reservoir conditions and to prevent any damage to the sample.

High precision pump is used to flow our fluids in the sample are used with an accuracy of 0.00001 ml/min and a pressure delta on 0.1 psi, and the cell is placed inside an oven to control the temperature. The test start with pumping 3% KCl brine from production direction at 0.5 ml/min, Continue pumping through the core until a constant flow rate till a stable ΔP pressure is achieved (at least 3 PV), this is Repeat the for at least two more rates then we heat the core holder to 275 F and when stabilized we repeat previous pumping procedure.

Then slick water is pumped from the opposite end of the core at a constant 1.0 ml/min rate for a total of 5 pore volumes, then Stop the pump and shut in the core sample for 18 hrs. at 270F keeping 2500 psi confining pressure on the core and after 18 hrs flow is Resumed with of the 3% KCl brine through the core in the production direction at 1 ml/min flow rate for a total of 8 pore volumes, then we set the injection rate to 3 ml/min for one pore volume, then 2 ml/min for one pore volume, and last 1 ml/min for one pore volume, and calculate the core permeability using Darcy equation with the slope method.

$$K(mD) = \frac{1000 * Slope * Viscosity * Length}{cross\ sectional\ area}$$

Interfacial tension

Samples are to be placed in in dish where its temperature will be monitored, a small metal mesh is cleaned and a torch is used to burn any residue, after which it will be installed in the IFT equipment, then it will be zeroed and then it will measure the surface tension multiple times, and correct with temperature, and then display the results showed in figure 5.

Results and Discussion

It can be observed many tests that is done with the additive compared to without it that there is an enhancement in many aspects FR testing, it is prominently apparent in the aging test where in 2 tests for the same mix, the one with the additive looks more clear and with less suspended residue that will decrease the regain permeability due to the plugging of available porosity (figure 2, figure 3, and figure 5), an improved phase separation of the friction reducer polymer after 24-hour aging and kept broken polymer particles suspended.

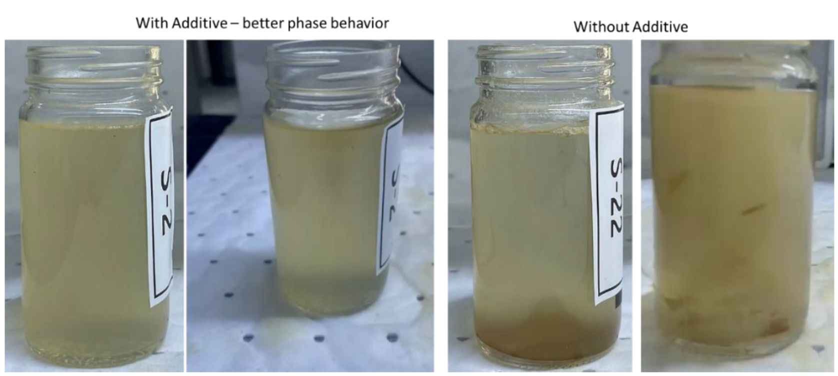


Figure 2—Sample 2 test 1– With and Without Performance Enhancer

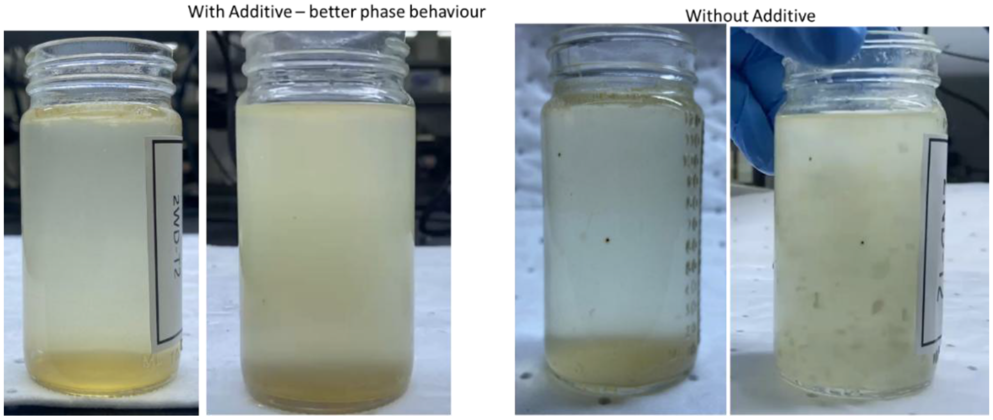


Figure 3—Sample 2 test 2– With and Without Performance Enhancer

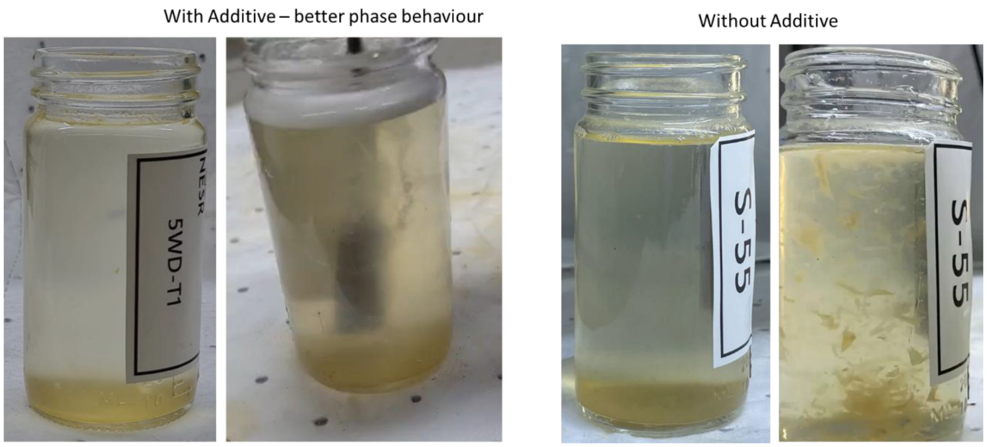
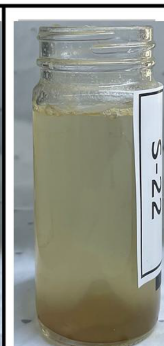


Figure 4—Sample 5 test 1 – With and Without Performance Enhancer

	With Additive	Without Additive
Surface tension measurement for the formulations with and without performance enhancer additive		
Surface Tension (mN/m) @ Temp of 20.6 oC	24.36	28.92
	24.38	29.11
	24.04	28.54
	24.15	28.78
Average Surface Tension (mN/m)	24.23	28.84

Significant decrease in surface tension

Figure 5—Surface tension – With and Without Performance Enhancer

The second test that show effectiveness of our additive is the surface tension for our aged slick water, it was reduced by from 28.8 to 24. mN/m showing an improvement of ~16% which is a large improvement in terms of phase behavior of broken slick water formulation after the frac job.
as for the regained permeability we can observe in table 2 that the regained permeability has increased by 7% compared to using only the original surfactant, This can be owned to the fact that the formation of mixed micelles between alcohol ethoxylate and silicone polyether (a performance enhancer) enhances surfactant properties and improves the phase behavior of the polymer This combined with the FR aggregation reduction, also helped reduce the impact the regain permeability by dispersing broken polymer in the brine. Due to synergistic micellar structure formation between two surfactant, high-temperature HVFR formulations tolerance was achieved.

Table 2—regain permeability results

Sample ID	Initial Permeability, mD	Post Permeability, mD	Retained Permeability, %
Without	152.557	123.122	80.71
With	150.689	131.626	87.35

Conclusion

To conclude, this study introduced a novel additive that enhance mixed micelle formation, refining polymer phase behavior and HVFR flowability. The additive increased regain permeability by up to 7% compared to conventional HVFR formulations and reduced permeability damage. This HVFR formulation can operate at high-temperature while minimizing phase separation of the friction reducer polymer after 24-hour aging and kept broken polymer particles suspended. The additive showed effectiveness in reducing the surface tension showing an improvement of ~16% which is a large improvement when taking in to account the low porosity and permeability formation, Overall, the novel additive significantly improved HVFR performance for high temperature and high salinity slick water applications.

The innovative chemistry that is incorporated in this formulation required extensive research and understanding of the structure-property relation to design optimized formulation.

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